

Time Slicer — Redesign Brief

For AI agent implementation | Flat Design Refresh

Overview

This document is a complete implementation brief for redesigning the `powerbi-time-slicer` custom visual. The goal is to replace the current mixed-style UI (chunky orange buttons, blue slider, pill badges) with a cohesive, flat design that feels at home in professional Power BI dashboards.

Repo: <https://github.com/Feltzem/powerbi-time-slicer>

Files to modify: `style/visual.less`, `src/visual.ts`

Style reference: `STYLE_GUIDE.md` (in this repo root)

Current State — What's Wrong

Element	Current problem
Reset button	Filled orange — too visually heavy, clashes with blue
Preset buttons	Pill shape with blue border — inconsistent with flat guidelines
Selected badge	Blue filled — dominates the hierarchy
Slider thumb	Default browser rendering — varies across OS/browser
Title ("Time Range")	Serif-ish default font, no visual hierarchy
Overall spacing	Uneven — elements feel loosely arranged
Colour palette	Three competing accent colours (blue, orange, green)

Target State — What to Build

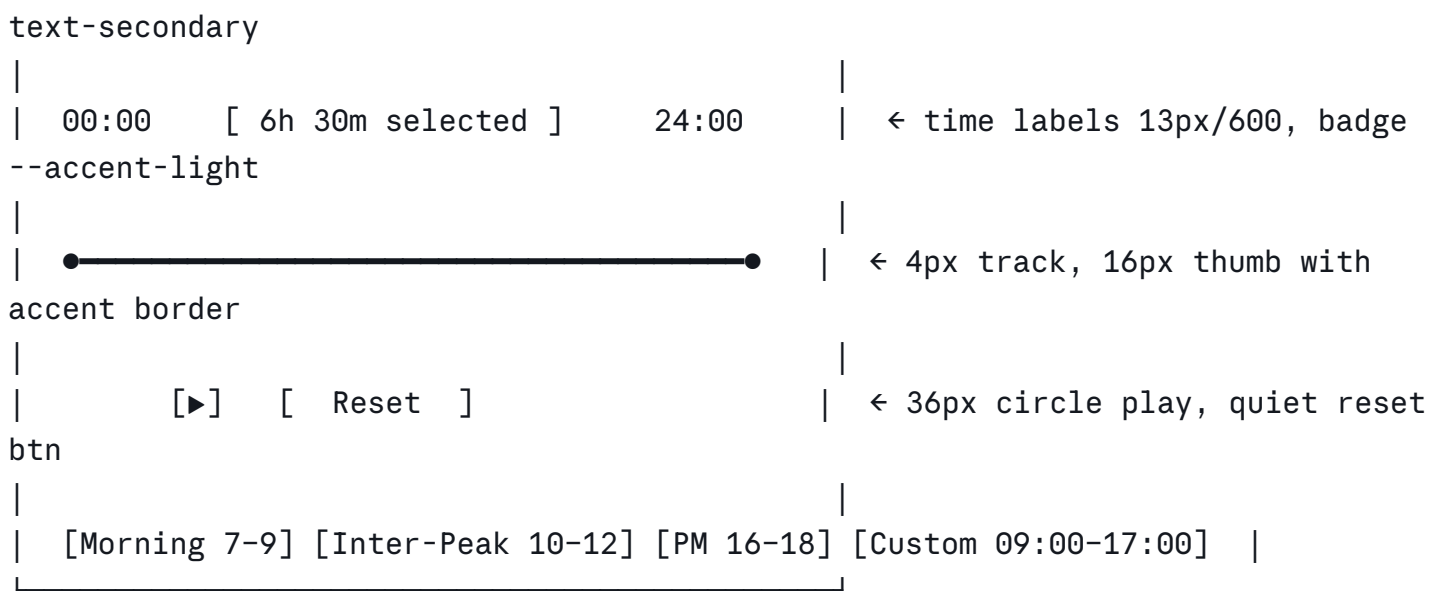
A single accent colour ( `#2563EB`), neutral greys for secondary elements, DM Sans typeface, and a clean 4px spacing grid throughout. See `STYLE_GUIDE.md` for all exact values.

Visual layout (top to bottom):

TIME RANGE

[label]

← 11px uppercase, --color-



Implementation Tasks

Task 1 — `style/visual.less`

Replace the entire stylesheet with the following structure. Use CSS custom properties (variables) declared on `:root` so Power BI theming overrides can work.

Variables block

```
less

:root {
  --color-accent:          #2563EB;
  --color-accent-light:    #DBEAFE;
  --color-accent-dark:     #1D4ED8;
  --color-bg:              #FFFFFF;
  --color-surface:         #F8FAFC;
  --color-border:          #E2E8F0;
  --color-track-bg:        #E2E8F0;
  --color-thumb:           #FFFFFF;
  --color-text-primary:    #0F172A;
  --color-text-secondary:  #64748B;
  --color-success:         #10B981;
  --color-reset:           #F1F5F9;
  --color-reset-text:      #475569;
  --font-family:           'DM Sans', 'Segoe UI', system-ui, sans-serif;
}
```

Container

```
less
```

```
.timeSlicerVisual {  
  font-family: var(--font-family);  
  background: var(--color-bg);  
  padding: 16px 20px;  
  display: flex;  
  flex-direction: column;  
  gap: 16px;  
  box-sizing: border-box;  
  min-height: 140px;  
}
```

Title

```
less  
  
.visual-title {  
  font-size: 11px;  
  font-weight: 600;  
  letter-spacing: 0.08em;  
  text-transform: uppercase;  
  color: var(--color-text-secondary);  
  text-align: center;  
}
```

Time labels row

```
less
```

```
.time-labels-row {
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 8px;
}

.time-label {
  font-size: 13px;
  font-weight: 600;
  color: var(--color-text-primary);
  letter-spacing: 0.02em;
  min-width: 40px;
}

.time-label--end {
  text-align: right;
}

.selected-badge {
  background: var(--color-accent-light);
  color: var(--color-accent-dark);
  font-size: 12px;
  font-weight: 500;
  padding: 4px 10px;
  border-radius: 6px;
  white-space: nowrap;
  transition: opacity 80ms ease;
}
```

Slider track + thumbs

```
less

.slider-container {
  position: relative;
  height: 20px;
  display: flex;
  align-items: center;
  padding: 0 8px;
}

.slider-track-bg {
  position: absolute;
  left: 8px; right: 8px;
  height: 4px;
```

```
background: var(--color-track-bg);
border-radius: 2px;
}

.slider-track-fill {
  position: absolute;
  height: 4px;
  background: var(--color-accent);
  border-radius: 2px;
  /* left and width set dynamically via JS */
}

/* Override native range input */
input[type="range"] {
  -webkit-appearance: none;
  appearance: none;
  position: absolute;
  left: 0; right: 0;
  width: 100%;
  height: 4px;
  background: transparent;

  pointer-events: none;

  &::-webkit-slider-thumb {
    -webkit-appearance: none;
    appearance: none;
    width: 16px;
    height: 16px;
    border-radius: 50%;
    background: var(--color-thumb);
    border: 2px solid var(--color-accent);
    cursor: pointer;
    pointer-events: all;
    transition: transform 100ms ease, border-color 100ms ease;

    &:hover {
      border-color: var(--color-accent-dark);
    }

    &:active {
      transform: scale(1.15);
    }
  }
}

&:focus::-webkit-slider-thumb {
  outline: 2px solid var(--color-accent);
  outline-offset: 2px;
```

```

    }

    /* Firefox */
    &::-moz-range-thumb {
        width: 16px;
        height: 16px;
        border-radius: 50%;
        background: var(--color-thumb);
        border: 2px solid var(--color-accent);
        cursor: pointer;
        transition: transform 100ms ease;

        &:active { transform: scale(1.15); }
    }
}

```

Controls row (play + reset)

less

```

.controls-row {
    display: flex;
    align-items: center;
    justify-content: center;
    gap: 8px;
}

.btn-play {
    width: 36px;
    height: 36px;
    border-radius: 50%;
    background: var(--color-accent);
    border: none;
    cursor: pointer;
    display: flex;
    align-items: center;
    justify-content: center;
    transition: background 150ms ease;

    svg { fill: #FFFFFF; width: 16px; height: 16px; }

    &:hover { background: var(--color-accent-dark); }

    &.playing {

```

```
background: var(--color-success);
}
}

.btn-reset {
background: var(--color-reset);
color: var(--color-reset-text);
border: none;
border-radius: 6px;
padding: 8px 16px;
font-size: 12px;
font-weight: 500;
font-family: var(--font-family);
cursor: pointer;
transition: background 120ms ease;

&:hover { background: var(--color-border); }
```

Preset buttons

less

```
.preset-row {
display: flex;
flex-wrap: wrap;
gap: 8px;
justify-content: center;
}

.preset-btn {
background: var(--color-surface);
color: var(--color-text-primary);
border: 1.5px solid var(--color-border);
border-radius: 6px;
padding: 6px 12px;
font-size: 12px;
font-weight: 500;
font-family: var(--font-family);
cursor: pointer;
transition: background 120ms ease, border-color 120ms ease, color 120ms ease;
white-space: nowrap;

&:hover {
```

```

background: var(--color-accent-light);
border-color: var(--color-accent);
}

&.active {
background: var(--color-bg);
color: var(--color-accent);
border-color: var(--color-accent);
font-weight: 600;
}
}

```

Task 2 — `src/visual.ts`

2a. Remove orange Reset button styling

Find any hardcoded colour values applied to the reset button element and remove them. The button's appearance is now fully controlled by `--color-reset` and `--color-reset-text` in the stylesheet.

2b. Apply DM Sans font

Add this `<link>` injection into the visual's DOM initialisation (inside `constructor` or `init`):

typescript

```

const fontLink = document.createElement('link');
fontLink.rel = 'stylesheet';
fontLink.href = 'https://fonts.googleapis.com/css2?family=DM+Sans:wght@400;500;600';
document.head.appendChild(fontLink);

```

If the visual runs in an offline/air-gapped environment, bundle DM Sans via `npm install @fontsource/dm-sans` and import it instead.

2c. Dynamic slider track fill

The filled portion of the slider track must be computed and applied in JS (since CSS `appearance: none` removes the native fill). Add or update a method:

typescript

```

private updateTrackFill(): void {
const min = this.sliderMin; // your existing min value
const max = this.sliderMax; // your existing max value

```



```

const max = this.sliderMax, // your existing max value
const start = this.currentStart;
const end = this.currentEnd;

const leftPct = ((start - min) / (max - min)) * 100;
const widthPct = ((end - start) / (max - min)) * 100;

this.trackFillEl.style.left = `${leftPct}%`;
this.trackFillEl.style.width = `${widthPct}%`;
}

```

Call `updateTrackFill()` on every slider `input` event.

2d. Play button icon swap

Use inline SVG rather than emoji/text characters for the play and pause icons so they scale cleanly:

typescript

```

const ICON_PLAY = `<svg viewBox="0 0 16 16"><polygon points="4,2 14,8 4,14" fill="
const ICON_PAUSE = `<svg viewBox="0 0 16 16"><rect x="3" y="2" width="4" height="

// Toggle in the play button click handler:
this.playBtn.innerHTML = this.isPlaying ? ICON_PAUSE : ICON_PLAY;
this.playBtn.classList.toggle('playing', this.isPlaying);

```

2e. Remove hardcoded colour overrides

Audit `visual.ts` for any `element.style.backgroundColor`, `element.style.color`, or similar inline style assignments that were previously needed to apply the orange/blue styling. Remove them — all colour is now handled through CSS variables and class toggles (`.active`, `.playing`).

2f. CSS variable theming from Power BI settings

When the user changes `accentColor` in the formatting pane, update the CSS variable on the container rather than applying styles per-element:

typescript

```

private applyTheme(): void {
  const accent = this.settings.timeSlicer.accentColor || '#2563EB';
  this.container.style.setProperty('--color-accent', accent);
  // Derive light/dark from accent if needed, or just update those too:
  this.container.style.setProperty('--color-accent-dark', this.darken(accent, 10)
}

```

Acceptance Criteria

When the redesign is complete, the visual should satisfy all of the following:

- ☐ No orange anywhere in the default theme
- ☐ Reset button is quiet — neutral grey, no border
- ☐ Slider thumb is a clean white circle with accent border (no browser default styling)
- ☐ Track fill is accent blue between the two thumbs only
- ☐ All buttons share the same 6px border-radius
- ☐ Preset buttons use outlined-active style (accent border + text, not filled) when selected
- ☐ DM Sans font is applied throughout
- ☐ Play button is a circular accent-blue button that turns green when playing
- ☐ Spacing is consistent (4px grid)
- ☐ No box-shadow on any element
- ☐ No gradients
- ☐ Selected badge is light blue pill (not bold filled blue)
- ☐ All interactive elements have a visible focus state for keyboard accessibility
- ☐ CSS variables are used for all colours (no hardcoded hex values in .less rules outside the :root block)
- ☐ Visual renders correctly at minimum 200px width

Out of Scope

The following are **not** part of this redesign task:

- Changing any Power BI data binding or filter logic
- Adding new preset buttons or changing their time ranges
- Modifying capabilities.json beyond what's noted in Task 2f
- Dark mode implementation (document as a future task)
- Animation of the slider playback

File Change Summary

File	Type of change
style/visual.less	Full rewrite — see Task 1
src/visual.ts	Targeted edits — see Task 2a-2f
STYLE_GUIDE.md	New file — design system reference (do not modify)